Transport and Endpoint as Free Parameters in Agentic Evaluation:

A Pre-Registered Study of Four Models Across Two Clouds

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Abstract

An agentic evaluation reports a number for a model, but that number is produced by an apparatus — a transport carrying tool calls, an endpoint serving them — named in no table. We measure how much of it belongs to the apparatus: four models, two managed clouds (Databricks and Amazon Bedrock), two transports, 45 held-out reverse-engineering tasks, 5,809 trials, analysis script frozen by hash with the hypotheses before any data existed (two of the ten tests were implemented a day later under a dated succession, when the data on disk could not inform either — §3).

Three results, in the order a reader should weigh them: two exploratory, one confirmatory. First, the variance components of an agentic measurement are specific to the apparatus that produced them, and that breaks study sizing. Inheriting a calibration from an earlier chapter on the same corpus, model and metric, ours did not mis-estimate a magnitude — it inverted which lever binds: 84% run-to-run noise and a floor of thirteen tasks became 0.5% (bootstrap interval $[0.0, 1.4]$ over 45 binaries) and a floor of 668. A designer who inherited it would not merely have been under-powered; they would have spent the budget on the axis that buys nothing.

Second, an endpoint can remove a model from an evaluation entirely, and we document *eight* mechanisms across eleven probed models, each with the endpoint's verbatim message. Six are reachable by a pre-flight probe — protocol refusal, deprecation, absent quota, disuse, jurisdiction, parallel calls — and two fire only in a conversational state a probe does not construct, so a harness certified healthy before collection can still lose a cell after the money is spent. This is a selection effect rather than a bias: the refused model has no row to correct. Removing one model from a four-model comparison moves the first-to-last spread by up to 47.6pp, against 9.83pp for the largest apparatus effect we measure.

Third, the confirmatory family resolves no effect at the available precision — none of the ten tests survives correction, and the three reaching nominal significance are the three least powered (21.0%, 30.9%, 35.8%). That is not a finding of no effect. We report the minimum detectable effect per contrast rather than one figure for the design, because the observed standard deviations span a factor of 23.7: resolving power is a property of the cell, not of the study.

Keywords agentic evaluation · tool calling · experimental validity · pre-registration · serving infrastructure · selection effects

Clinical trial number: not applicable.

1 Introduction

An agentic evaluation reports a number for a model. That number is produced by an apparatus: a transport that carries tool calls between the model and the harness, an endpoint that serves the weights, a set of tools with a particular return format, and a decoding configuration. The model's name appears in the table. The apparatus does not.

This paper measures how much of the number belongs to the apparatus rather than to the model, and finds that the question has two answers of different kinds. The transport moves a number, which is a bias: it can be bounded, and a reader who knows its size can correct for it. The endpoint deletes a row, which is a selection: it cannot be bounded, because the model that was refused has no line in the table to correct.

1.1 The confound this chapter removes

The preceding chapter of this programme (Author(s) withheld, 2026) measured a 10.4pp drop when one model moved from native function calling to a textual protocol. That chapter is a companion manuscript under double-anonymous review, so a reader cannot obtain it and we do not ask anyone to take its numbers on trust: both figures this paper draws from it recompute from a snapshot of the exact cells they need, released here with their per-turn logs. On a single infrastructure that number admits two readings no further collection on that infrastructure can separate — a property of the model, or of that provider's shim — because model and infrastructure are perfectly confounded when there is only one endpoint.

Serving the same nominal model from two managed clouds separates them: if the difference persists it belongs to the model, if it moves it belongs to the infrastructure. That cell — same nominal model, two commercial endpoints, transport manipulated as a paired factor, on a multi-turn downstream task — is what this study contributes as a design, and to our knowledge it has not been run.

1.2 Contributions

The order below is the one in which we would have a reader weigh them, not the one in which they were planned.

1. **A power calibration does not transfer between apparatuses**, even holding model, corpus and metric fixed: inherited from the preceding chapter, ours did not mis-estimate a magnitude, it inverted which of two levers binds (§6) — a claim about how experiments of this kind should be sized, and the finding we did not go looking for.

2. **Eight mechanisms by which a serving endpoint removes a model from an evaluation before it produces a score** — six reachable by a pre-flight probe, two that fire only in a conversational state a probe does not construct — each with the endpoint's verbatim message (§5). The split is the operational finding: a harness certified healthy before collection can still lose a cell after the money is spent. Existing taxonomies classify *faults*, things that stop happening once repaired; these are refusals by a healthy endpoint behaving as configured, so they persist for a whole study (Pandey and Singh, 2026). For the one any single-call pre-flight reports healthy we give the constructed probe that catches it first.

3. **Two properties of the apparatus usually assumed rather than measured**: temperature zero is not determinism, and the same nominal model is not equally stable on two clouds — which makes score stability a property of the serving stack rather than of a decoding parameter (§6).

4. **A pre-registered** $4 \times 2 \times 2$ **design** — four models, two named managed clouds, two transports, 45 held-out binaries, 8 runs each — with the analysis script frozen by hash *together with* the hypotheses before any data existed, one documented exception covering two of the ten tests (§3), and the confirmatory outcome reported with the power that

produced it: no test survives Holm, and the three reaching nominal significance are the three least powered. We report the minimum detectable effect per contrast, because the observed standard deviations span a factor of 23.7 and no single band describes them.

5. **An artifact** carrying every raw row including the invalidated batches, the per-turn session traces, the frozen documents with their hashes, and the amendment log (§9).

1.3 What this paper does not claim

Not that one transport is better: we measure that the choice moves the number, not which choice is right. Not semantic equivalence: pass-rate is a lower bound with a floor we measure. Not that the eight mechanisms are exhaustive — each is an existence proof with its own denominator, never a rate. And not that the effect sizes travel: they belong to one task family, four models and two clouds. What we do claim is narrower and harder to escape: an evaluation that does not report its transport is not reproducible from its method section, whatever the task.

2 Related Work

We position this work against its neighbours and state the delta at the width the evidence supports.

2.1 Measurement bias, and the harness as its current object

The shape of our finding is old in systems research and new only in the evaluation of agents: that a measurement depends on the stack under it is unremarkable in performance work, where declaring the configuration is a norm. What agentic evaluation lacks is that norm. The argument has been made for the harness as a whole (Zhang et al., 2026; Yao et al., 2026); we are downstream of it and supply the measurement it lacks — what the undeclared choices are worth, one axis at a time.

Four lines of work bound this one, and the artifact expands each with the figures we take from it. **Apparatus variance is established methodology elsewhere**: rigorous performance measurement has long required reporting the configuration and treating run-to-run variation as first class (Georges et al., 2007; Iosup et al., 2011), and measurement bias from unreported setup choices was demonstrated two decades ago (Mytkowicz et al., 2009) — what is missing here is not the method but its application. **Non-determinism at nominal temperature zero** has been measured for repeated sampling (Atil et al., 2024) and traced to the batch size a server happens to be running (He and Thinking Machines Lab, 2025) — the class of mechanism our §6.7 finds diverging by a factor of 2.3 in input tokens. **The endpoint is already known to be a service rather than an artifact**: a measurement study of hosted open-weight APIs finds one model differing across providers in protocol support, error semantics, context capacity and throughput (Li et al., 2026), which is our T5–T8 axis arrived at independently. **And the transport axis is visible to the field, in no fixed direction**: a leaderboard ranks every model in a native and a prompted mode (Patil et al., 2025); on one model the prompted variant is *ahead* by 6pp (bfc, 2026; bfc); and a third transport, tools as typed code stubs rather than JSON, matches or exceeds native calling on 11 of 14 (Patel et al., 2026). Three transports, three directions. **And the task family has its own literature**: reconstruction from stripped binaries is active (Tan et al., 2024; Jiang et al., 2025), semantic equivalence under transformation has been studied on the corpus we reuse (De Tomasi et al., 2025), test-based equivalence has known limits (Liu and Wang, 2020), and the reproducibility of published results built on commercial models has been assessed directly, with model deprecation as a leading cause of failure (Angermeir et al., 2026).

2.2 The transport: known to move the number, in no fixed direction

The nearest thing to a counterexample is a public leaderboard ranking every model in both a native and a prompted mode: the axis is not invisible to the field. It reports the two modes side by side rather than crossing them with the serving endpoint, and on one model finds the textual variant *ahead* by 6pp — the opposite direction to what seeded this work (bfc, 2026; bfc), which is why we treat it as a constraint on our claim rather than support. Designing on an effect size estimated from earlier data is known to miss its target power (Albers and Lakens, 2018), and non-determinism at nominal temperature zero has been measured for repeated sampling (Atil et al., 2024); what this case adds to both is a failure that reverses rather than drifts, at a rate differing by cloud at fixed model identity.

The reporting obligation is already settled and we do not argue for it: community guidelines for empirical work with language models require the harness to be described when it goes beyond bare API calls (Baltes et al., 2026). Our contribution is not the rule but the measurement it lacks — how much the undeclared choices move, and that its threshold is set too high.

2.3 The endpoint: a unit of measurement, and a unit of selection

Holding weights, decoding and hardware fixed while varying only the inference engine produces measurable differences in output, which establishes the mechanism we extend: if an engine moves a number at fixed weights, a managed platform choosing engine, version and scaffolding without disclosing any of them can move it too, and a reader has no way to know by how much (Pape et al., 2026).

Removal also breaks reproduction downstream, and its scale has been measured on one public arena's own roster: of 243 models, 205 had been silently deprecated against 47 marked as such, at rates of 87.8% for open-weight and 80% for proprietary models (Singh et al., 2025). That audit names the mechanism — "Chatbot Arena may be forced to deprecate a model when a provider no longer supports it via its API" — and counts the outcome after the fact, aggregated over a roster; we measure the refusals themselves, verbatim and case by case, before a score exists.

2.4 Where the failure taxonomies stop

The nearest neighbour to our census is a taxonomy of failure modes in multi-provider serving infrastructure, classified by origin layer and detectability (Pandey and Singh, 2026). Two of its five layers already reach failures occurring before generation — network and transport covers "TCP timeouts, TLS handshake failures, DNS resolution issues", governance and cost covers "rate-limit counters that leak and never recover" — so the distinction we draw is not about *when* the failure lands.

It is about whether anything is broken. Every entry in that catalogue is a *malfunction* — an operator could in principle repair it, and the leaking counter we just quoted is a bug precisely because it should recover and does not. The mechanisms of §5 are the other kind: an endpoint behaving exactly as configured, refusing by policy, with nothing for anyone to fix. That is why they persist for the duration of a study. We develop that distinction, and what it implies for the taxonomy, in §5.7 where the mechanisms are on the page.

3 Method

3.1 The task

Each trial gives a model one stripped x86-64 binary, decompiled with a static analysis server, and asks it to reconstruct compilable C source. The candidate is compiled in a pinned container and executed against the unit tests of the original program, and the score is the fraction that pass.

Pass-rate is a *lower bound* on semantic equivalence and we report it as such throughout: a candidate correct but differing in an untested behaviour scores below one, and one wrong in an untested behaviour scores above its true worth. We use it because it is executable and adversarially hard to game, not because it is tight.

3.2 Pre-registration

The hypotheses, the test family, the falsifiers, and *the analysis script* were frozen by content hash before a single row of data existed. Registered reports have been available in this field since 2020 (Ernst and Baldassarre, 2023); what we add is the second half of that freeze.

Two of the ten tests are an exception we state here rather than let a reader find it in the artifact, because the exception is the more informative case. T9 (heterogeneity across models) and T10 (transport $\times$ infrastructure interaction) were declared in the family at the freeze but emitted a non-computable p, and were implemented eleven hours later under a dated succession document written *before* the change. Collection had begun, so the freeze does not cover them strictly, and their guarantee is a weaker but checkable property: neither could be informed by the data that existed — three cells were on disk, all of a *single* model, while T9 compares the transport effect *between* models and T10 needs both transports on both infrastructures, one of which had not started. Both entered Holm at the $m = 10$ fixed before any data existed, so the correction they face was not chosen after seeing them. Fixing the hypotheses while writing the analysis after seeing the data leaves intact every degree of freedom pre-registration exists to remove: which test, on which subset, with which exclusion. Every subsequent change to a frozen file is documented before the change and checked by a guard, and the register is in the artifact.

3.3 The two transports

The manipulated variable is how a tool call travels between the model and the harness. Both transports offer the same tools, the same schemas, and the same twelve turns.

Native. The request carries a `tools` field. The model emits a structured tool-call object, which the runtime parses; results return in the role the provider's API defines for tool output.

Textual. The request carries *no* `tools` field. The system prompt defines a line protocol: the model ends its reply with a single line `TOOL_CALL:` followed by a JSON object naming the tool and its arguments. The harness parses that line and returns the result as an ordinary user message.

Two consequences bound what the contrast means, and both belong before any number. First, the native transport admits more than one call per turn where the text protocol admits one by construction.

Table 1: Batching under the native transport, per model. The text protocol is fixed at one call per turn by construction, so the confound is present only where this rate exceeds one.

model	calls/turn	turns > 1	contrast affected
haiku	1.438	38%	T3
sonnet	1.333	33%	T4
gpt-oss	1.000	0%	—
llama	1.000	0%	—

The confound is structurally available to every model and binds only where the model uses it, which two of four do not at all. Second, *a turn that produces prose without a tool call is*

Table 2: The six questions, what answers each, and its status. Confirmatory items map onto hypotheses frozen before collection; exploratory items are reported as such throughout.

	question	answered by
RQ1	By how much does the transport of a tool call move the measured score, at fixed model and infrastructure?	H1, tests T1–T4 *(confirmatory)*
RQ2	By how much does the serving endpoint move it, at fixed model and transport?	H2, tests T5–T8 *(confirmatory)*
RQ3	Is the transport's effect constant across models, and do transport and endpoint interact?	H3 and H4, tests T9–T10 *(confirmatory)*
RQ4	By which mechanisms does an endpoint remove a model from an evaluation before it produces a score?	the census, §5 *(exploratory)*
RQ5	How much of the transport difference is the loss of call batching rather than the change of format?	the constrained-native arm, §6.5 *(exploratory)*
RQ6	Does a power calibration transfer between apparatuses that share model, corpus and metric?	§6 *(exploratory)*

not a failure: it is a turn without a call and is recorded as such, since treating it as a terminal answer would end the trajectory at turn one and score the protocol rather than the model.

3.4 What counts as a measurement

One predicate decides whether a row is a measurement, and collector and analysis use the same one — two definitions of validity, one per half of a pipeline, diverge silently. It has two clauses, and we state them here because every number in this paper is computed over the rows they admit: a row is a measurement unless the endpoint reported an infrastructure failure, or the harness itself raised. Rows the predicate rejects are kept in the released data with the reason attached — they are the subject of §5, not attrition — and the artifact states the failure that motivated each clause.

4 Experimental Design

4.1 Research questions

Six questions organise what follows: three answered by the pre-registered family, three by evidence labelled exploratory wherever it appears. They expound a design frozen before collection — no hypothesis, test or comparison is added, and the family stays at $m = 10$ (§4.5).

RQ4 through RQ6 were not all asked at the outset: RQ4 was pre-registered as an exploratory census, while RQ5 and RQ6 are questions the apparatus raised while it ran. What protects them is not that they were anticipated but that each is answered by a quantity a reader can recompute — verbatim endpoint responses, an arm with a stopping criterion fixed in advance, and two calibrations from two apparatuses.

4.2 The grid

Four models, each served by both Databricks and Amazon Bedrock under two transports: sixteen cells. We name the platforms rather than abstract them — the venue is single-blind, and the mechanisms of §5 quote the endpoints verbatim, so a reader recognises the platform from the message whatever we call it and can re-issue each probe against the platform that produced it. Every cell runs the same 45 held-out binaries eight times, so the design plans 5,760 trials while the primary re-collection delivers 5,809 and the replication batch 5,805. The three numbers differ for one reason: a run returning an infrastructure failure is logged as a record and *re-run*, so a cell can end with more valid measurements than the 360 it planned. The surplus — 49 above plan and 45 — is not a bonus but the count of runs an endpoint refused or dropped, which §5 treats as the subject rather than as attrition, and one cell ends *below* plan at 44 binaries of 45 for the reason given in §5.5. Turn budget, temperature and output cap are identical in every cell and recorded per row rather than assumed.

Table 3: The grid. Every cell is the same corpus under a different apparatus.

models	4 (two open-weight, two proprietary)
infrastructures	2 (Databricks, Amazon Bedrock)
transports	2 (native, textual)
cells	$4 \times 2 \times 2 = 16$
binaries per cell	45, held out, frozen by hash
runs per binary	8
trials	5,760 planned, 5,809 measured
turn budget	12
temperature	0.0

4.3 Why we say *nominal* model

Throughout we write *nominal model* rather than *the same weights*. Two managed clouds advertising the same open-weights model may serve different quantisations, different runtime versions, or route to a different variant under load, and neither exposes enough to settle it. Verifying byte-identity is not possible from the outside, so we do not claim it: what we hold fixed is the model *identity a user selects*, which is exactly the thing a published evaluation names in its table.

This is not a hedge, it is the object of study. If two endpoints selling the same name behave differently, that difference belongs in the method section of every paper that names the model and not the endpoint — whether the cause is the weights, the runtime, or the routing.

4.4 Hypotheses, and why two of them are disjunctive

We ask whether the transport moves the score at fixed infrastructure (H1), whether the infrastructure moves it at fixed transport (H2), whether the transport's effect is constant across models (H3), and whether transport and infrastructure interact (H4).

This matters for how the result is read. An interval that excludes effects above some magnitude is not the same claim as no difference, and we never write the second when the data support only the first.

4.5 The test family, and why its size does not shrink

Ten tests were pre-registered as one family with Holm step-down. The family size is fixed. It does not shrink when a cell turns out to be inconclusive, uninteresting, or unexecutable, because removing a member after seeing the data lowers the thresholds of every survivor — a correction that grows more permissive exactly as one learns which tests one would like to pass.

One consequence is deliberate and worth naming: the family includes contrasts nobody disputes, and those contrasts consume correction budget. That is the price of fixing m before the data exist, and it is cheaper than the alternative.

4.6 Six defaults inherited from the preceding chapter

Six constraints are inherited from the preceding chapter, each traceable to a defect that produced a clean and false number: unwrapped candidate source, functions in native order, symbol

Table 4: Mechanisms observed while constructing the roster. Each is an existence proof with its own denominator, not a rate. **Level** says where the decision that removes the model is taken, and it is the column that decides whether a reader can do anything about it: *vendor* holds for everyone using that platform; *region* depends on where the request originates; *tenant* depends on the account's own history and quota, so two teams on one platform can see different rosters; *conversational state* fires only inside a session and is invisible to any check made before one starts.

mechanism	level	evidence
protocol refusal	vendor	`400`: multi-turn tool calls unsupported, on 43 of 45 binaries
deprecation	vendor	`ServiceModelDeprecated … since 06/13/2026` — while a second cloud still serves the same nominal model
absent quota	tenant	the catalogue advertises the deployment type; the quota system has no row for it, and creation is refused
disuse	tenant	*"marked by provider as Legacy and you have not been actively using the model in the last 30 days"*
jurisdiction	region	*"Access to Meta Llama models is not allowed from unsupported countries, regions, or territories"* — while two other models of the same family run
parallel calls	conv. state	`400 UnsupportedToolUse`: *"This model does not support more than one tool call at this time"*

tables stripped, a reasoning-only turn not recorded as a zero, a price guard that refuses *before* the provider call, and the token-limit parameter sent under the one name the endpoint accepts. None is a discovery — each is a default a competent implementer would set, which is what makes the list worth publishing — and the artifact states each with the defect it answers.

5 RQ4 *(pre-registered as exploratory)*: What the Endpoint Removes Before a Score Exists

This section reports the result that does not depend on a p-value: an endpoint either returned the message we quote or it did not. Building the sample required probing which models each infrastructure would serve, and that probe returned six mechanisms by which an endpoint removes a model from an evaluation. Two more appeared from inside the confirmatory arm, where a probe cannot reach them: eight in total, none sought. The split is the practical finding: a pre-flight reaches the first six, not the last two.

We report the six from the probe separately from the two found by collecting, because the distinction is the practical finding. A pre-flight check reaches the first six; the last two fire only in a conversational state that a check does not construct, and one of them is visible on one transport and invisible on the other.

5.1 The six a probe reaches, with the endpoint's own words

Table 4 is not a taxonomy derived from documentation: every row is a message an endpoint returned to a request we made, and the artifact carries the request that produced it.

5.2 The one a probe reports healthy

Five of the six are found by asking the endpoint an ordinary question. The sixth — the refusal of more than one tool call in a turn — is not: it needs a history in which the model has already emitted two calls in one assistant message, a state no single request constructs. The useful consequence is that the state is nonetheless *constructible*, and hand-assembling that history is a probe: send it and the endpoint answers before any collection has been paid for. The mechanism was found by collecting and only then reduced to something findable by asking.

5.3 A seventh case, inside the confirmatory arm

One more constraint belongs here rather than in the threats, being the same phenomenon observed from inside our own design. The protocol we pre-registered ends each trajectory with a turn that offers no tools, and on Bedrock that turn *cannot be expressed*: the API requires a tool configuration whenever the history contains a tool call, and its tool-choice parameter admits “auto”, “any” and a named tool — but not “none”.

The exposure was not where it first appeared. The broken rows carried an infrastructure-failure flag and were already excluded from every mean; what survived was structural. A trajectory reaches the final turn with a tool call in its history only if it has *not* already submitted through a tool, so the runs that completed were exactly those that had submitted early — a second selection effect, inside the axis this study exists to measure cleanly. We discarded that batch and re-collected the cell against a corrected client. It is the sixth mechanism’s twin: a constraint a pre-flight probe reports as healthy, because it fires only in a conversational state a probe does not reach.

5.4 Four further cases, in the artifact

Four observations belong to the census and not to a paper of this length, and the artifact carries each with its verbatim evidence: how the roster was assembled and what each denominator is; the same model returning *two different refusals* one minute apart, depending which of the vendor’s own identifier conventions is used; one model comparable on one cloud, refused by a second, fully functional on a third; and a service control policy permitting invocation while denying enumeration. The last carries the only obligation that transfers — under such a policy the roster cannot be recovered from the account and must be declared by explicit enumeration, which is what §4 does — and is a property of our account rather than of the platform, labelled as such.

5.5 An eighth case: the endpoint rejects the output of the model it serves

The re-collection produced one more, the sharpest of the set because neither party is malfunctioning. The open-weights model emits tool calls in its own chat format, which uses special tokens to separate reasoning channels; the serving layer does not strip that token from the tool name, so the name reaches the API as `decompile_function<|channel|>commentary` — eleven occurrences with that exact value. The API then validates the name it was just handed:

```
ValidationException: Value at
'messages.10.member.content.1.member.toolUse.name'
failed to satisfy constraint: Member must satisfy
regular expression pattern: [a-zA-Z0-9_-]+
```

and rejects the entire request. Not the turn — the trajectory. *The endpoint refuses the output of the model it is itself serving*, and no party is at fault: the model emits its native format, the client sends a valid request, the validator does its job. The serving stack does not close the loop between the format of the model it hosts and the validation of its own API, and retrying does not help — seven valid runs on one binary took fifteen.

5.6 How far a removal moves a published comparison

That a refused model has no row to correct is an argument, and an argument does not say how much a reader’s conclusion changes. The census documents real cases — a Llama variant deprecated on one platform while another serves it, a second blocked by jurisdiction while its siblings run — so someone evaluating on one endpoint reads a table with rows missing and does not know which. We bound the consequence on the four models where a score exists on both clouds, removing one model at a time and recomputing the comparison the way a leaderboard reports it.

The *order* survives: all pairwise orderings between the two clouds agree before and after every removal. The *scale* does not. Removing the lowest-scoring model collapses the first-to-last spread from 54.3pp to 6.7pp on one cloud and from 47.3pp to 10.4pp on the other, and across all eight removals the largest change is **47.6pp** — against the −9.83pp that is the largest transport effect this study measures, derived in §6. A single missing row moves what a benchmark reports as "the distance between the systems compared" by roughly five times what the apparatus effect moves it.

This is a *lower* bound and the direction of the extrapolation is worth stating: with four models there is little room to move, and the same mechanism on a twenty-model leaderboard has more, not less. What our data support is the sign and the order of magnitude, not a coefficient.

5.7 Why this is the part that is new

The closest neighbour is a taxonomy of failure modes in multi-provider serving, classified by origin layer and detectability (Pandey and Singh, 2026). Our eight mechanisms do not sit inside it, for a reason sharper than a gap in coverage: that taxonomy classifies *faults*, each entry something that stops happening once repaired. The question none of its layers asks is the one deciding whether a model appears in a benchmark at all — *will this endpoint serve this model identity?* — so what we offer is a missing layer rather than an empty cell, with Table 4 as its evidence: eight instances, each with the verbatim message of the endpoint that produced it. We do not claim nobody has catalogued serving failures; we claim the catalogues begin one step after a model can vanish from a study. The artifact carries the layer-by-layer comparison.

6 Analysis and Interpretation

The census establishes that an endpoint can remove a model before a score exists. This section asks how far the surviving numbers move, and answers with the resolving power attached.

All numbers here come from the re-collection: an independent execution of the full $4 \times 2 \times 2$ design on workspaces isolated per cell, comprising 5,809 valid measurements. That this batch, rather than the original one, would carry the primary analysis was fixed in writing before it produced a row (§8); the original collection is reported alongside it as a replication, and the two agree on every criterion declared in advance.

6.1 Which lever binds is a property of the cell

The variance of a paired difference decomposes into within-binary noise, which shrinks with more runs per binary, and heterogeneity between binaries, which shrinks only with more binaries. The two are not interchangeable, and Table 5 shows the choice is not a property of the axis.

For T3 and T6 more runs buy nothing: 668 and 380 binaries against the 45 collected.

These are estimates on 45 units, so we report their intervals. Bootstrapping the binaries — the unit the contrast is paired on, 5,000 replications at a declared seed — gives T3's noise share as 0.5% $[0.0, 1.4]$ and K_∞ as 668 $[265, 1136]$, and T6's as 9.1% $[4.1, 21.6]$ and 380 $[126, 744]$. The interval on K_∞ is wide and its lower end is still six times the 45 binaries

Table 5: Variance decomposition. K_∞ is the number of binaries needed to resolve 3pp with unlimited runs per binary. †: the estimated noise exceeds the observed variance, so the variance model does not fit that cell and its decomposition is not reported — the bootstrap clamps the residual to zero in 86% and 69% of replications respectively.

	axis	SD	% noise	K_∞
T3	transport	0.2774	0.5%	668
T6	endpoint	0.2189	9.1%	380
T4	transport	0.1409	20.6%	137
T1	transport	0.2002	46.1%	189
T2	transport	0.0916	61.0%	28
T5	endpoint	0.1292	93.1%	10
T7	endpoint	0.0117	not decomposable†	
T8	endpoint	0.0585	not decomposable†	

collected, which is the operational point: the conclusion that more runs buy nothing does not depend on the point estimate. Neither interval reaches the 50% that would make the noise share indistinguishable from half.

Where the estimator is misspecified, we say so rather than offer two readings. The rows marked † have estimated noise exceeding the observed variance — not a finding about those cells but a variance model that does not fit them, since the decomposition assumes the two conditions' runs are independent given the binary and clamps the residual at zero when that fails. The bootstrap makes the failure countable: the residual is clamped in **86%** of replications on T7, 69% on T8 and 43% on T5, so for those contrasts the lower bound of K_∞ is zero by construction rather than by evidence, and we report no decomposition for T7 and T8. On T1, T3 and T6 — the contrasts that carry the finding — clamping occurs in at most 1% of replications.

What the metric's granularity does and does not explain

A decomposition of variance is only as trustworthy as the measurement it decomposes, and ours is coarse: pass-rate over five unit tests gives a run *six* values, 53% of the 5,809 rows sit at an extreme and 33% of the 720 binary-by-cell means land exactly on 0 or 1. A floor- and ceiling-heavy measurement is on its own sufficient for a large between-binary spread, so it is a rival account of Table 5 and not only of the inversion that follows.

Two conditions bound the account, and both are stated so that a reader can check them. It would have to explain why the same metric on the same corpus yields 0.5% noise here and 84% in the collection this design inherited its sizing from — granularity is a property of the metric, and a property the two share cannot by itself produce a difference between them. The second condition is narrower than the first and we state it as such: resampling binaries bounds the sampling variance of the estimate, not a measurement bias, so the interval $[0.0, 1.4]$ on T3 does not by itself exclude a floor-and-ceiling pattern uniform across the corpus — such a pattern would reproduce in every resample. The first condition carries the argument on its own, and neither is closed by these data alone.

The operational consequence is in §7.5: a design resting on a variance decomposition should size the number of tests per task the way it sizes the number of tasks, and report the share of runs at floor and ceiling beside the decomposition. We did the second and not the first.

What T3 and T6 share is not the axis but the pairing of largest observed standard deviation with smallest nominal p: where the effect looks largest it also varies most between binaries, which is a weaker claim than a large average effect.

6.2 RQ6 *(exploratory, arose during execution)*: the power calibration inherited from the previous chapter inverted

The pre-registration states, from the previous chapter's data, that for haiku "84% is run-to-run noise and the floor is $K = 13$". On the same nominal contrast here, the noise share is 0.5% and the floor is $K = 668$.

This is a reversed mechanism rather than a magnitude error, and it survives three independent changes: of estimator, which widens it to 130.7% noise and $K = 0$; of apparatus and workspace isolation, where the two collections give 0.1%/705 and 0.5%/668; and the other high-variance contrast, 9.3%/487 against 9.1%/380. The artifact carries each recomputation.

The 84% comes from the batch in which a validity flag failed, and the three robustness checks above do not touch that. Crashed rows were averaged as zeroes there (§3), and the released snapshot lets a reader reconstruct both computations. We claim no direction for that defect: independent reconstructions disagree, and on that corpus the corrected estimator is misspecified in the sense of §6.1, so the quantity one would compare against is clamped rather than estimated. What the reconstruction does establish is that the inherited 84% is the *defective* computation, which is why the inversion is not read off the comparison: it is established inside this chapter by two collections of our own, 0.1%/705 and 0.5%/668 on the contrast that carries it.

One caveat is removed by none of the three: the chapters differ in apparatus as well as estimator, and exposure to the largest information leak of §8 was asymmetric between arms in the earlier chapter — 100% of native runs invoked the affected tool against 86.7% of textual runs — so the comparison does not isolate the estimator. The inversion is larger than 13pp of differential exposure makes plausible as a sole cause, but that is an order-of-magnitude argument, not an exclusion. What does not depend on the comparison is this chapter's own decomposition: the 0.5% noise share and the 668-binary floor are computed on our data alone and under one estimator, so the finding that the binding lever is a property of the apparatus stands whether or not the earlier number is right.

6.3 RQ1–RQ3 *(confirmatory)*: a negative registered result, and the resolving power that is the reportable quantity

This subsection reports a *negative* pre-registered result and reports it as such: hypotheses, test family and analysis were fixed before the data existed, the family does not resolve an effect, and the outcome is stated before the exploratory findings that did produce something. What it contributes is not the absence of an effect but the *resolution* at which the absence was established, which differs by a factor of 23.7 between contrasts sharing one design.

Of the ten tests fixed in the pre-registration, *none* passes its Holm threshold at $m = 10$ (Holm, 1979). The smallest p is 0.0191 against a threshold of 0.0050. No contrast places its effect outside the ±3pp falsifier band with a confidence interval that excludes it.

Part of that null follows from the family's fixed composition. Two of the ten tests are not independent of the other eight, so Holm at $m = 10$ divides the smallest threshold by ten rather than eight — 0.0050 instead of 0.0063. T6's p of 0.0191 misses both, so no conclusion turns on the choice here, though on a study whose smallest p fell between them it would; we keep $m = 10$ because the pre-registration fixed it and because it is the conservative direction, not as the only defensible count. The artifact carries both series.

That answers RQ1–RQ3 in the direction of "not at this resolution", and the resolution is the reportable quantity: an evaluation that cannot separate 4pp from zero is not free to call two transports equivalent. The outcome does not rest on the t's normality assumption either — Appendix A repeats the family under a distribution-free sign-flip test and no test passes Holm there.

Table 6: The ten pre-registered tests, on the re-collection (isolated workspaces). δ is the paired difference per binary, over 45 binaries of 8 runs each. **Sign convention**: for transport contrasts δ = textual − native, so a negative value means the textual protocol scores lower; for endpoint contrasts δ = Bedrock − Databricks. MDE is the effect this contrast could have resolved at 80% power given its own observed standard deviation, in percentage points — the column exists because it varies by a factor of 23.7 across contrasts of one design. Power is computed against the pre-registered MDE of 4.87pp using the *observed* standard deviation and the non-central t — the distribution the test actually uses. A normal approximation would report 0.0–1.9 points higher, which in a paper reporting its own under-powering would err in the flattering direction. ‡The p column is the *exact* Student series, not the approximation the frozen script emits: the approximation understates p in eight cases of eight — for T6, 0.0155 against 0.0191 — always in the direction that makes an effect look more significant. Both series are in the artifact, the distribution-free series is in Appendix A, and no test passes its Holm threshold under any of the three. ¶T5 is the one contrast whose p rests on K=44 while its δ and interval rest on K=45: one binary returned seven valid runs of eight on one endpoint, the frozen script averages it and the exact-p script requires a full complement. We print both rather than harmonise them, because harmonising would mean choosing a rule after seeing what it does to a pre-registered number.

	contrast	δ	95% CI	$p^{\ddagger}$	power	MDE
T6	llama, endp.	−7.94	[−14.5, −1.4]	0.0191	30.9%	9.14
T3	haiku, transp.	−9.83	[−18.2, −1.5]	0.0218	21.0%	11.59
T1	gpt-oss, transp.	−6.56	[−12.6, −0.5]	0.0334	35.8%	8.36
T9	heterogeneity	—	—	0.1011	—	—
T2	llama, transp.	+2.22	[−0.5, +5.0]	0.1107	93.7%	3.82
T5¶	gpt-oss, endp.	+3.26	[−0.6, +7.1]	0.1159	69.6%	5.40
T7	haiku, endp.	+0.22	[−0.1, +0.6]	0.2095	100.0%	0.49
T8	sonnet, endp.	−0.94	[−2.7, +0.8]	0.2844	100.0%	2.44
T10	interaction	−4.12	[−10.6, +2.3]§	0.3448	—	—
T4	sonnet, transp.	−1.00	[−5.2, +3.2]	0.6363	62.1%	5.88

We report the family outcome together with the achieved power, never one without the other, because the three contrasts with nominal $p < 0.05$ are also the three least powered — and reporting power at all is the exception here: 103 software engineering experiments were found systematically below accepted norms (Dybå et al., 2006), a decade after the concern was raised as “missing and misunderstood” (Miller et al., 1997), and documented again for language-model evaluation (Card et al., 2020).

Table 6 and Fig. 1 are the whole confirmatory result. T3 reproduces the transport difference that motivated this chapter at −9.83pp — different apparatus, isolated workspaces, three information leaks closed (§8) — and T6 shows 7.94pp between two clouds serving the same nominal model at fixed transport. Neither survives correction for multiplicity, and both sit below a third of the power the design was sized for. All three are true together.

§ The frozen script prints no interval for T10; this one is computed outside it with the *model* as the unit of replication ($n = 4$, t_3). Treating the 180 binary-level pairs as independent would give a narrower interval and a wrong one: they are four groups of 45.

6.4 RQ1–RQ3 *(confirmatory)*: the resolving power is per contrast, not a single band

The pre-registration sized the design with a pooled standard deviation of 0.1062 and a conservative 0.1167, both calibrated on the previous chapter. The observed standard deviations of the paired differences span 0.0117 to 0.2774 — a ratio of 23.7 — so a single minimum detectable effect describes the design rather than the data. Computed per contrast, the MDE ranges from 0.49pp (haiku across endpoints) to 11.59pp (haiku across transports), and *five of*

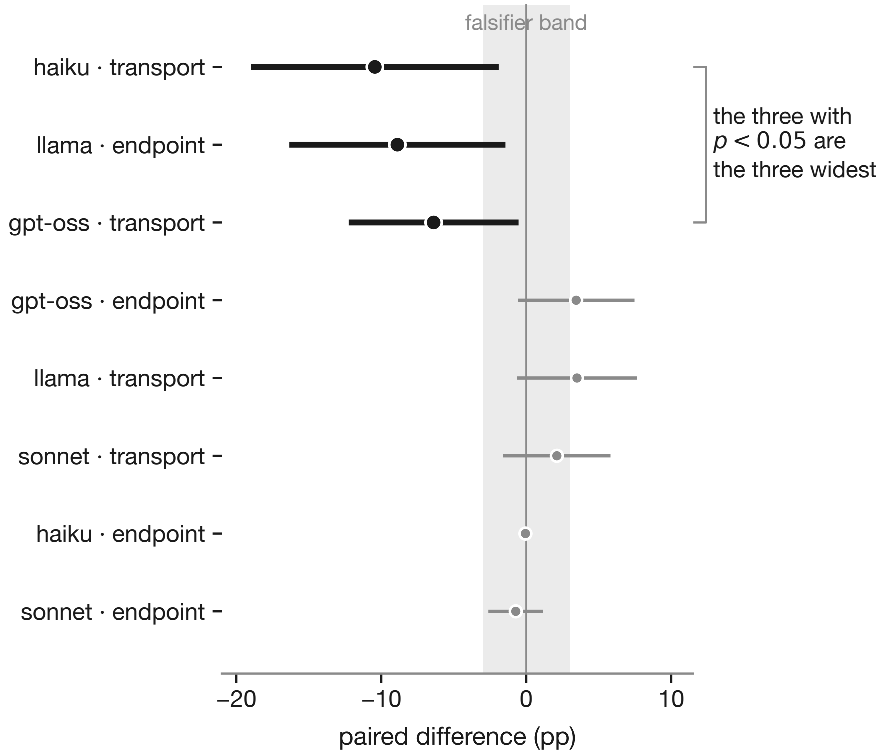

Figure 1: The ten contrasts, ordered by nominal p. Filled and heavier lines are the three with $p < 0.05$ uncorrected; none survives Holm correction. They are also the three widest intervals in the study — the signature of a winner's curse, not of a larger effect. Dotted lines mark the pre-registered ±3pp falsifier band.

Table 7: Constraining the native transport to one call per turn. Paired on binaries, $K = 45$, eight runs per binary. The arm is exploratory and does not enter the corrected family.

model	full	constr.	diff.	95% CI
haiku-4-5	0.747	0.754	+0.8pp	$[-2.6, +4.2]$
sonnet-4-5	0.793	0.809	+1.6pp	$[-1.3, +4.4]$

eight contrasts exceed the 4.87pp the design assumed.

A null result here therefore reads: no contrast places its effect outside the band its own variance permits. It does *not* read as equivalence — $K = 45$ does not power an equivalence test at 3pp, which would need 99–119 binaries, and this was declared before the data existed.

6.5 RQ5 *(exploratory, arose during execution)*: evidence against call batching as the explanation

The pre-registered transport contrast moves two things at once on the models that batch: the format of the call and the number a turn admits — haiku issues 1.438 calls per turn against sonnet's 1.333. A declared confound is not a bounded one, so we collected the arm that separates them: native transport forced to *one call per turn*, all else held fixed. It is exploratory, does not enter the family of ten, and its stopping criterion was fixed before it ran.

Removing batching does not lower the score: both differences are *positive* and both intervals contain zero (Table 7), against a transport effect of −9.83pp on the same model and endpoint. The direction is what the arm supports — a mechanism cannot explain a decrease it does not produce — and that inference needs only the sign of the two differences, which is why it survives the arm's missed validity band (§6.6): the band governs how much the constraint bit, not which way the score moved.

The magnitude does not follow, and we do not claim it. "Not within an order of magnitude" would require the arm to have removed batching at the rate the band specified, and it did not. What remains is narrower and still useful: the loss of call batching is not a plausible *principal* account of T3, and no reading of this arm establishes that the remainder is the protocol and nothing else.

Two further cautions belong to the arm and are in the artifact: that the constraint does not remove batching alone, so what it bounds is an upper limit rather than an estimate; and that the arm covers one infrastructure and the two models that batch at all, since on the other two the rate is exactly 1.000 calls per turn and there is nothing to remove.

6.6 RQ5, second finding *(exploratory)*: one refusal reconfigures the model for the rest of the run

The arm produced a result we were not looking for, and it is sharp. Across 1,084 trajectories *every* first turn issues more than one call and is constrained; after the first refusal the rate collapses to 0.2%, eight turns out of 4,457. The model does not learn the constraint from the tool schema, which does not encode it — it learns it from one rejection, and the lesson holds for the remainder of the trajectory.

The arm falls outside its pre-declared validity band, and that governs what it licenses. The band required 33–38% of turns to carry a constrained call, estimated from the unconstrained batching rate; the observed figure is 19.7%, outside it and not marginally. Under the rule fixed in §4, the ablation therefore does not license a statement about the *size* of the batching effect and we make none. Why the band was wrong is itself a measurement, and it is sharper than a missed target: the band assumed a batching rate constant along the trajectory, and the collapse above fixes the achievable share at essentially one constrained turn per trajectory. Of the 5,541 turns that carry a call, 1,084 are first turns and every one of them is constrained; eight more are constrained later, for 1,092 in total, or **19.7%** — which is the structural ceiling of 19.6% plus the eight turns that still batched after a refusal. The band's 33–38% was never reachable under this constraint, so its premise was wrong rather than the data.

The two statements are not interchangeable. *The arm is not evidence about magnitude* — that reading requires the band it missed. *It is evidence about direction*: both constrained cells score no lower than unconstrained, which is incompatible with the loss of batching explaining a 9.83pp drop whatever its size, since a mechanism cannot explain a decrease it does not produce. A sign survives a failed magnitude check; an effect size does not. It also explains an asymmetry in the main comparison: under the textual protocol the option to batch never exists, while under constrained native it exists for one turn and the model abandons it. A transport is not only a format a call travels in — it is a channel that teaches.

6.7 What differs between two clouds serving one model identity

T6 measures 7.94pp between two platforms serving the same model identifier, and §8 states that the weights cannot be verified identical — which says nothing about *how* the two runtimes differ. Four quantities in the raw rows are determined by a serving stack rather than a model, and they do not concord.

The two platforms hand the model **2.3 times** the input tokens for the same task, corpus and twelve turns, while emitting output at within 6% of the same rate. Throughput agreeing while input diverges points away from hardware and towards what each stack *sends* — prompt scaffolding, tool-schema serialisation, history truncation — and which of the three is not visible

Table 8: T6's two cells, native transport, 360 runs each. Median with first–third quartile; the ratio is Bedrock over Databricks. Turn budget and corpus are identical by construction.

	Databricks	Bedrock	ratio
input tokens	20,562	8,853	0.43
latency per run (s)	13.4	8.6	0.64
output tokens	448	303	0.68
candidate characters	598	544	0.91
turns	12.0	12.0	1.00
output tokens per second	33.2	35.1	1.06

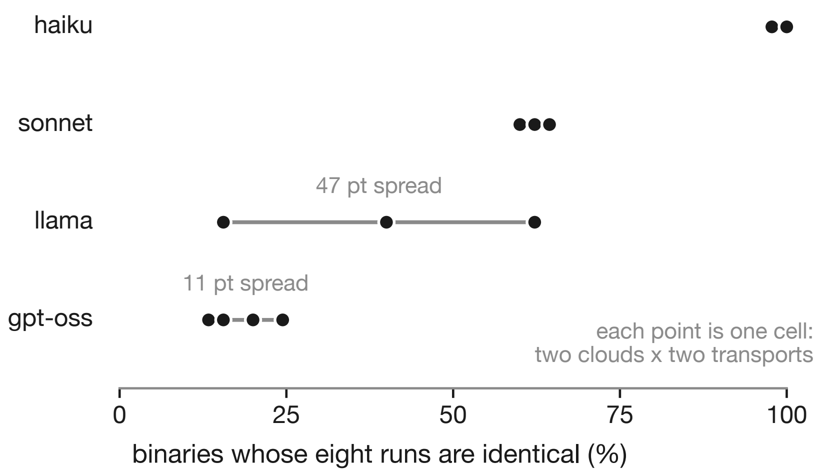

Figure 2: Determinism at temperature 0.0: the share of binaries whose eight runs return an identical pass-rate, per model, with one point per cell. The spread within a model is the distance between its clouds and transports — the same nominal model is not equally stable on different infrastructures.

from outside. This changes the status of T6's confound rather than its size: declared before, documented now, with something in the two stacks differing by a factor of 2.3 on the quantity the preceding chapter identified as a free parameter in its own right (Author(s) withheld, 2026).

6.8 Temperature zero is not determinism, and not equally so

Two properties of the apparatus are usually assumed and turn out to be measurable. *First, temperature zero is not determinism.* All cells ran at temperature 0.0, and across the eight runs of the same binary in the same cell haiku returns an identical pass-rate on 91–100% of binaries against gpt-oss's 20–24%, with 56% of binaries over the study yielding eight identical runs (Fig. 2).

Second, *the same nominal model is not equally stable on different clouds*, and on 45 binaries the sampling supports that claim in exactly one of eight comparisons: llama on the native transport returns identical scores on 24.4% of binaries at one endpoint $[14.2, 38.7]$ and 66.7% at the other $[52.1, 78.6]$, Wilson bands not touching. The other seven overlap and are not used as evidence. One clean case is what these data support, on the same model and axis where T6 shows the largest endpoint effect in the study, and the original collection isolates the same comparison.

6.9 What the metric buys, measured

Pass-rate has a floor reachable without reconstructing anything, and we measured it rather than assuming it: the 53 runs that called *no* tool average 0.1132 against 0.6378 for the 5,756 that called at least one. The floor exists and does not dominate — the metric is a lower bound with a measured floor, and the artifact carries the distribution.

6.10 Declared covariates

Two confounds were measured before the confirmatory numbers were read and neither is corrected for, since correcting a pre-registered design after the fact is the freedom pre-registration exists to remove. *Calls per turn* is the one the ablation arm of §6.5 isolates, and it binds on two models of four (§3). *Material acquired* is the second: replaying what each transport actually requested, the native arm asks for more functions per binary in *all eight* cells, most strongly on llama at one endpoint (−1.16 functions, SD 0.90), against six of eight originally. We do not order that per-cell agreement against the size of the endpoint effect — with four models the ordering is indistinguishable from noise — and the artifact carries both covariates in full.

6.11 Scope

The effect sizes are circumscribed twice over and both bounds are part of the result. They come from *one* family of tasks — decompiled C binaries scored by unit tests — on four models and two clouds, so T3's −9.83pp is a measurement on this corpus rather than a quantity transferable elsewhere: a reader who needs a number for their own setting should take the method and the variance decomposition, not the point estimate. And every statement above concerns *this* text protocol — one call per turn, one line of the response — not textual tool calling in general, about which a leaderboard reporting both modes finds the textual variant *ahead* by 6pp on one model, against the direction that seeded this work.

7 Discussion

7.1 A null confirmatory result that is not a null finding

Ten pre-registered tests, none surviving correction — and the reason is measurable rather than mysterious. The design was sized on a standard deviation borrowed from the preceding chapter (Author(s) withheld, 2026), and that borrowing is where the study earns its keep: the quantity borrowed turns out not to be a property of the model, so no care at design time would have produced a different number of binaries. Where the pre-registration recorded 84% run-to-run noise and a floor of thirteen binaries, the same nominal contrast here gives 0.5% and 668. Not an underestimate — an inversion, and the lever the previous chapter said would work is the one that buys nothing here.

This is the finding we did not go looking for, and §6.2 places it against the literature on borrowed effect sizes. The consequence for a designer belongs here: inheriting that calibration would not merely have left them under-powered, it would have sent their budget to the axis that buys nothing. The rival account for the decomposition — that the granularity of the metric produces the spread on its own — is bounded in §6.1 by the two conditions it would and would not have to explain, with the figures for each.

7.2 Why the census carries the weight

The confirmatory arm and the census differ in what it takes to overturn them. Every number in §6 is a point estimate with an interval that a larger study could move; nothing in §5 is an estimate, since an endpoint either returned that message or it did not and the artifact carries

Table 9: What an agentic evaluation should declare. The first four sharpen requirements that already exist (Baltes et al., 2026); the last two have no counterpart in current checklists.

field	what it means
transport	how a tool call is encoded: native function calling, a textual protocol, typed code stubs — and whether more than one call per turn is admitted
endpoint	which service served the weights, not only the model name
model and version	the identifier the endpoint accepts, including any regional prefix
configuration	temperature, token limits, turn budget, seed where honoured
models excluded	which models were considered and not measured
reason for each	the endpoint's own message, verbatim, or the policy that applied

the request that produced it. A referee who disbelieves the census can re-run the probe; one who disbelieves the contrasts has to re-run 5,809 trials.

7.3 When to discard a batch, and when not to

Two batches were invalidated and re-collected rather than patched, and the rule separating them from the one we kept is reusable. **Discard when the contaminating channel is systematic and asymmetric between the compared arms; report and bound it otherwise.**

The larger invalidation — 2,390 measurements — meets that test exactly: the algorithm's name reached the model through three separate channels and the two arms encountered one at different rates, so left in place it would have turned part of the transport effect into a measure of how often each transport stumbled onto the answer. No downstream analysis separates a contaminant that correlates with the manipulated variable.

The shared working directory (§8) fails the same test in the other direction: non-directional noise, some thirty times in the original batch's 5,805 measurements, with no asymmetry between arms. Discarding it would have been a gesture rather than a correction, so we bounded it, probed it, and re-collected on isolated workspaces. Two practices made the rule applicable rather than rhetorical: channels were enumerated per channel and not per intention — the third was found because we listed every place the model could read — and both batches remain in the artifact with their recomputation.

7.4 What an evaluation should report

What follows amends an existing checklist rather than proposing a new one: the community guidelines already require the harness to be described once it goes beyond bare API calls (Baltes et al., 2026), and each field below either sharpens that requirement or extends it to one they do not name. Each costs a line in a method section, and Table 9 states them as fields because a recommendation phrased as prose is one a reader agrees with and does not apply. The first carries the argument of §6.7: a model name does not identify a measurable system, the same nominal model on two clouds being two apparatuses that differ in score stability as well as in score.

7.5 Five propositions that should outlive this task

The effect sizes here belong to one task family, four models and two clouds, and we claim nothing beyond that. What we offer beyond it are five propositions about how experiments of this kind behave, each instantiated by this study rather than proved in general.

1. The transport belongs beside temperature in a method section.

2. An endpoint can produce missing experimental units, whose cost §5.6 bounds.
3. Variance components can be apparatus-specific, so an inherited calibration must be recomputed on one's own data as soon as the first cell exists.
4. A pre-flight must exercise realistic conversational states: two of our eight mechanisms fire only in a state a single-call probe never reaches.
5. Metric granularity limits what a variance decomposition can say. We report the floor and ceiling shares beside ours, and did not size the tests per task.

The artifact carries each with its argument.

8 Threats to Validity

Each threat is reported in the same four parts: what it is, how large it is, what we did about it, and what remains. The threats an artifact reviewer would find anyway are here too, with the same four parts and no different treatment.

8.1 Construct validity: what pass-rate measures

Threat. Pass-rate is a lower bound on semantic equivalence, and it has a floor: a candidate written without consulting the decompiled program can still pass some tests, so part of any score may be pattern-matching rather than reconstruction.

Quantification. The floor is measurable inside the study rather than assumed, and §6 reports it: the runs that called no tool at all average 0.113 against 0.638 for those that called at least one.

Residual. The floor exists and does not dominate, and both halves of that sentence carry a number. Pass-rate remains a lower bound and is reported as one throughout; no claim in this paper requires it to be more.

8.2 Construct validity: what a transport carries

Threat. The transport contrast is a change of *format* and also a change in how much context reaches the model. §6.7 treats a factor of 2.3 in input tokens between two clouds as evidence that the two stacks differ; the same quantity differs between the two transports, on the axis this study exists to isolate.

Quantification. Median input tokens per run, native over textual, across the eight cells where both arms exist: 2.89 and 2.47 on gpt-oss (Databricks, Bedrock), 2.34 and 1.02 on llama, 1.38 and 1.37 on haiku, and 0.75 and 0.73 on sonnet. The native arm receives more in six cells of eight and fewer in two, so the divergence has no fixed direction and its extreme, 2.89 on T1, exceeds the 2.3 the endpoint subsection reports. T1 is the cell where this matters most: gpt-oss issues exactly one call per turn, so it is the contrast with no batching confound to absorb the difference.

Diagnostic. This is a component of the transport, not a leak into it. Under native calling the tool schemas travel in a `tools` field and results return in the provider's tool role; under the textual protocol both travel as ordinary messages. Two encodings of one conversation do not cost the same number of tokens, so no version of this contrast can hold the count fixed while changing the format — the count is part of the format. The ablation arm of §6.5 separates batching from format; nothing here separates format from context volume.

Residual. The transport effects of §6 are therefore effects of a transport as deployed rather than of a serialisation syntax holding all else equal, and a reader sizing their own study should read them that way — the same lesson the ablation arm returns from the other side.

A transport is not only a format a call travels in: it is a channel with its own budget, and it teaches.

8.3 Internal validity: a working directory shared across cells

Threat. Candidates were compiled in a directory keyed on the program and run index and nothing else, so sixteen cells shared 360 directories under concurrent collection and a run compiling another cell's candidate would be scored normally with no flag raised. The key omits the cell, so no contrast is structurally exempt.

Quantification. Thirty pairs of rows from different cells share a program and run index within two seconds; across the full grid 843 of 5,805 measurements (14.5%) have a row from another cell that close, and all sixteen cells are affected. The artifact gives the mechanism, the write-to-read window and the probe these data admit — which reassures about nothing, its sign being the opposite of a collision's and vanishing once model identity is controlled for.

The designation of the re-collection as primary basis is *argued* rather than *certified*, and the difference is a number. The re-collection ends at 5,880 *rows*, of which 5,809 are measurements under the predicate of §3; the criteria script's timestamp falls when 112 of those rows existed — **1.9%**, all from a *single* cell of sixteen — and no criterion of the four is computable from one model — so their content could not have been informed by their outcome, which is checkable in the released rows. The files enter version control the next day, when 73.4% of the rows existed, and a modification time can be rewritten. The pre-registration proper (§3) carries the stronger form: committed 22 hours before the original collection's first row, and that precedence verifies.

Diagnostic and mitigation. Because a probe that cannot settle the question is not a mitigation, the working directory now carries the cell and the experiment was collected again on workspaces isolated per cell. Two quantities separate there. *Crowding* — a row from another cell within two seconds — is a property of sixteen concurrent cells and survives isolation: 843 measurements originally, 451 after. *Shared compilation paths* is the mechanism itself, and it does not: 30 originally, *zero* under 451 concurrent measurements that would have exposed it. Removed by construction, not observed absent.

The four frozen criteria, and their outcome. All four are met. *Sign agreement*: six of eight paired contrasts keep their direction, against a threshold of six. *Family outcome*: no test passes its Holm threshold in either batch, under both the frozen approximation and the exact Student series — four series, one answer. *Interval coverage*: seven of eight re-collected effects fall inside the original 95% interval, the exception being T7, whose original interval is $[-0.2, +0.1]$ because that cell is almost noiseless — a demanding target rather than a disagreement about the effect. *Variance decomposition*: the noise share that inverted the power calibration reproduces, $0.1\%/K_\infty = 705$ originally against 0.5%/668, with the second high-variance contrast at 9.3%/487 against 9.1%/380. The criterion asked for agreement within an order of magnitude and got it on the quantity that carries the finding.

The effect sizes move by up to 3.1pp (T4, the smallest and least stable), and no conclusion of the chapter depends on one of them individually. What the re-collection establishes is narrower and more useful than agreement: the confirmatory result does not rest on an apparatus whose sharing behaviour cannot be audited.

Residual. Whether a collision affected the original batch is not decidable from the released data — the working directory is overwritten by design and is an audit trail of nothing — so that batch is neither invalidated nor defended, and the criterion this programme applies, with the two batches it did invalidate, is in the artifact.

8.4 Conclusion validity: the guards, and their perimeter

Threat and residual. The analysis is protected by content hashes and an append-only convention, and neither guarantee is as wide as its name: the hash chain covers six files,

Table 10: The same eight contrasts measured twice: the original collection, and an independent re-collection on workspaces isolated per cell. Differences in percentage points; the last column states whether the re-collected effect falls inside the original 95% interval. The criteria and their thresholds were written before the comparison could be computed; their timing is stated precisely in §8, where it is weaker than the pre-registration's.

	model	orig.	re-coll.	in CI
T1	gpt-oss-120b	−6.4	−6.6	yes
T2	llama-3.3-70b	+3.5	+2.2	yes
T3	claude-haiku-4-5	−10.4	−9.8	yes
T4	claude-sonnet-4-5	+2.1	−1.0	yes
T5	gpt-oss-120b	+3.4	+3.3	yes
T6	llama-3.3-70b	−8.9	−7.9	yes
T7	claude-haiku-4-5	−0.1	+0.2	no
T8	claude-sonnet-4-5	−0.7	−0.9	yes

three of them analysis scripts, and not the other thirty in that directory; and the append-only rule is enforced by a hook that stops known destructive commands but not an ordinary shell redirection, so the real guarantee there is the committed history a reader can check. We state the perimeters rather than let the names imply they coincide, and the artifact enumerates what each covers.

8.5 External validity: what the design does not settle

Threat. The design manipulates transport and infrastructure at a fixed *nominal model identity*, and we use the weaker phrase deliberately: it does not verify that a nominally identical model is byte-identical across two providers, and neither provider exposes enough to settle it.

Quantification. The divergence matters rather than being hypothetical: on the one axis where the Wilson bands separate, the same nominal model returns identical scores on 24.4% of binaries at one cloud against 66.7% at the other (§6).

Residual. We report the rates and decline to attribute them, because attributing them would require knowing what each provider serves. The scope of every effect size reported here is one task family, four models and two clouds, as the pre-registration bounds it.

9 Conclusion

Four models, two managed clouds and two tool transports were run against a pre-registration and an analysis script frozen together before any row existed, save two tests documented in §3. The confirmatory result comes from the independent re-collection on workspaces isolated per cell — 5,809 measured trials — designated the primary basis in writing before it produced a row; the original batch of 5,805 sits beside it as a replication, and the two agree on all four criteria frozen in advance. None of the ten pre-registered tests survives Holm correction, and the three contrasts reaching nominal significance are the three least powered in the study — reported together with the achieved power, since power determines what a p-value can mean.

The design was mis-sized, and the reason is itself the finding: the calibration inherited from the preceding chapter did not misjudge a magnitude, it inverted which lever binds. Where that chapter found 84% run-to-run noise and a floor of thirteen binaries, the same nominal contrast gives 0.5% and 668. A calibration is a claim about an apparatus, and it does not travel.

What does not depend on a p-value is the census: six pre-collection mechanisms remove a model from an evaluation, each with the endpoint's own words, plus two conversational-state failures that appeared only while collecting. The split is the practical finding — a pre-flight probe reaches the first six and *cannot* reach the last two, so a harness certified healthy can still lose a cell after the money is spent, and existing taxonomies of serving failures begin one step later.

Two properties of the apparatus usually assumed turned out to be measurable and to vary. Temperature zero is not determinism: one model returns identical runs on 91–100% of binaries and another on 20–24%. And in the one comparison of eight where the sampling bands separate, the same nominal model returns identical scores on 24.4% of tasks at one endpoint and 66.7% at the other, which makes score stability a property of the serving stack rather than of a decoding parameter.

The transport moves a number, and that is a bias: bounded, correctable, already visible in a public leaderboard. The endpoint deletes a row, and that is a selection: the model it refused has no line to correct. Evaluations report the first inconsistently and the second not at all. Declaring the transport, the endpoint, and the models each endpoint refused costs three lines in a method section, and without them a published agentic number does not identify the system it describes.

Declarations

Funding

This work received no specific grant from any funding agency. Inference costs were borne by the author's institution and are itemised per arm in the replication package, so the cost of each number is auditable beside the number.

There are no acknowledgements: the author is the sole contributor, and this statement is therefore already separate from any.

Ethical approval

Not applicable. The study involves no human participants and no personal data. It measures commercial inference endpoints under their ordinary terms of use, at a request rate below the published limits of both platforms, and reports the responses those endpoints returned.

Informed consent

Not applicable. The study has no human participants, so there is no one from whom consent could be sought. The subjects of measurement are four commercial inference endpoints, and what was collected from them is the text they returned to documented requests.

Author contributions

Lorenzo De Tomasi is the sole author and carried out the whole of the work: conceptualisation, the pre-registration and its frozen analysis plan, the harness for both transports, data collection across both clouds, the analysis scripts, the interpretation, and the writing of the manuscript. There are no other authors.

Data availability statement

The replication package contains the 45 stripped binaries and their test suites, the decompilations, the harness with both transports, every raw result row including the two invalidated batches, the per-turn session traces, the frozen pre-registration and analysis script with their content hashes, and the dated succession documents accounting for every change to a frozen file. Each number in this paper is produced by a script released with it, re-running from the released rows; no figure is drawn by hand.

The two numbers this paper takes from the preceding chapter are re-runnable here: the deposit carries a snapshot of exactly the cells they need with their per-turn logs, so both the calibration inversion and the tool-exposure asymmetry recompute without leaving the artifact. It does not carry the rest of that chapter, and says so.

Three commands regenerate everything a reader might want to check, and they are the ones we ran:

`python3 analysis/numeri_paper.py` — every number in prose
`python3 analysis/tabella_principale.py` — Table 6
`python3 analysis/tabella_principale.py --varianza` — Table 5

The first carries its own known-answer check: pointed at the original collection it must reproduce the numbers this paper reported before the re-collection, and exits non-zero otherwise — which is what makes the new series trustworthy rather than merely recomputed, and what caught a ratio reported as 77 whose value is 76.3. Each collection is selected by the same three environment variables (`C2_RESULTS`, `C2_PATTERN`, `C2_PREFISSO`); setting only one yields an index built on one batch and queried with the keys of the other, which returns zero rather than raising.

The package is openly available at `https://github.com/lodetomasi/transport-endpoint-agentic-eval` and will be deposited on a permanent archive with a DOI on acceptance. Cloud workspace and profile names, absolute paths and one internal account identifier are replaced by placeholders, because infrastructure names belong to an employer rather than to a study, and `REDAZIONE.md` lists every substitution; nothing else is redacted, the venue being single-blind. Redaction changes the content hash of a frozen file and this paper quotes those hashes, so the deposit carries its own recomputed table alongside that document, with the integrity guard run inside the deposit and passing there. Two further properties are checkable rather than asserted: the result directory is append-only, invalidated batches moved and annotated with the recomputation beside each rather than deleted; and every change to a frozen file carries a document written before the change, declaring the previous and the new hash, with a guard rejecting any divergence no document accounts for.

Conflict of interest

The author declares no competing interests. The two serving platforms measured in this study were selected because the institution already held accounts on both; neither vendor was contacted, informed, or involved at any stage, and neither reviewed this manuscript.

Reporting checklist

The study follows the community guidelines for empirical studies involving language models (Baltes et al., 2026): model identifiers, versions and configuration are reported per cell (§3); the system and harness design is described and released; per-turn session traces are published for every run rather than summarised; and two of the four models under evaluation are open-weight, so that their scores can be independently re-run on infrastructure the reader controls. The guidelines' requirement that we sharpen — reporting the transport — is the subject of the paper rather than only its practice.

Use of language models in this work

Language models are the object of study here, and they were also used as a tool while conducting it — to draft and review code and prose — which the community guidelines ask to be declared separately from their role as subject (Baltes et al., 2026). No measurement, no number, and no decision reported in this paper was produced by that use: every quantity comes from a committed script run against committed rows, and the guards described above exist precisely because an author who accepts a generated number without recomputing it has no way to notice when it is wrong.

A Distribution-free sensitivity for the pre-registered family

The pre-registered test is a paired t over 45 binaries, which assumes approximately normal paired differences. On this metric they are not: pass-rate over five unit tests takes six values, 53% of rows sit at an extreme, and the paired difference of two such means is discrete with short tails. If a test assuming no distribution reaches the same conclusion, the assumption was not carrying the result. For paired differences the exact permutation test is the sign flip — under the null the sign of each binary's difference is exchangeable — and with 2^{45} assignments we sample 50,000 at a declared seed and report the Monte Carlo error of the p itself, a sampled p being an estimate.

Table 11: Sign-flip permutation against the pre-registered Student's t. Holm applies the same fixed $m = 10$ as the pre-registered family.

	p Student	p permutation	MC error	difference
T6	0.0191	0.0135	0.0008	−0.0055
T3	0.0218	0.0228	0.0011	+0.0010
T1	0.0334	0.0319	0.0012	−0.0014
T2	0.1107	0.1194	0.0023	+0.0087
T5	0.1159	0.1205	0.0023	+0.0046
T8	0.2844	0.3267	0.0033	+0.0423
T7	0.2095	0.5032	0.0035	+0.2937
T4	0.6363	0.7754	0.0030	+0.1391

No test passes Holm under either series, so the family outcome does not depend on the distributional assumption and the objection closes without altering anything pre-registered.

Two contrasts disagree by more than the Monte Carlo error, informatively: T7 moves from 0.21 to 0.50 and T4 from 0.64 to 0.78, and both are the contrasts with the smallest observed variance, where the t divides by a standard error the permutation test never uses. That the t reports smaller p exactly where variance is smallest is the mechanism rather than a coincidence, and it supports the reading of §6: T7's narrow interval reflects absence of noise rather than power. This analysis is not pre-registered and does not replace the frozen one — had the two diverged on the family outcome the reportable conclusion would still be the pre-registered one, with the divergence declared.

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